MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three types of integrally lighted indicators for indicating the flow rate of engine fuel in pounds per hour (pph) for three specific flow ranges. The indicator is a hermetically sealed synchro/servo receiver with a redundant counter/pointer display. The indicator may provide a voltage output signal proportional to the fuel flow rate and contains special damping provisions to attenuate small oscillatory transmitter output signals.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

Type	Integral Lighting	Flow Rate (PPH)
EFU-22A/A	Red	0-80,000
EFU-22A/A-2	White	0-80,000
EFU-22A/A-7	White	0-80,000

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods of
MIL-E-5400	-	Electronic Equipment, Aerospace, General Specification for
MIL-C-14806	-	Coating, Relfection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25467	-	Lighting, Integral, Red, Aircraft Instrument, General Specification for
MIL-C-26500	-	Connector, General Purpose, Electrical Miniature, Circular, Environment Resisting, General Specification for
MIL-L-27160	-	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	-	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-T-38144	-	Transmitter, Rate of Flow, Fuel

STANDARDS

MILITARY

DOD-STD-100	-	Engineering Drawing Practices
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interterence Characteristics, Measurement of
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	-	Dissimilar Metals
MS24265	-	Connector, Receptacle, Electrical Single Hole Mount Miniature
MS24266	-	Connector, Plug, Electrical Straight, Miniature
MS27034	-	Connector, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature, Hermetic Solder Mount
MS33558	-	Numeral and Letter, Aircraft Instrument Dial, Standard Form of
MS33585	-	Pointer, Dial, Standard Design of Aircraft Instrument
MS33639	-	Case, Instrument, Clamp-Mounted, Aircraft

(Copies of specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

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3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and test (see 4.4 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of indicator except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4.1 Reclaimed materials. There is no exclusion to the use of reclaimed materials.

3.3.5 Toxic and corrosive fumes. The indicator shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form toxic or corrosive alkali will not be liberated.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Lubricant. Silicone lubricants shall not be used without prior approval of the contracting activity.

3.4 Design and construction.

3.4.1 Function (all indicators). All indicators shall include all of the circuitry and mechanism necessary to convert a three-wire synchro signal from the applicable fuel flow transmitter into a dual pointer-counter presentation of fuel flow rate. The indicators shall contain a receiver synchro control transformer (Clifton Precision part no HTC-8-A-4/L086, or approved equivalent) to work with the synchro transmitter (Clifton Precision part no TGC-11-E-6) in the fuel flow meter. For an increasing flow rate, the direction of rotation of the transmitter synchro shaft is clockwise as viewed from the shaft end. The indicators shall provide an auxilliary voltage output signal, when specified, which shall be proportional to the indicated fuel flow rate.

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3.4.1.1 EFU-22A/A and /A-2 indicators. The requirements for these two indicators shall be identical except for lighting. The indicators shall work with the TRU-63/A transmitter conforming to MIL-T-38144. The signal from the transmitter is linear and corresponds to 1° per 500 pph. The rotor of the synchro transmitter will rotate over a linear range of 160° for a corresponding fuel flow rate of zero to full scale. With an input excitation voltage of 115V 400 Hz, the auxiliary voltage output signal shall be 10.0V for a full scale input of 80,000 pph. The auxiliary output signal shall be provided for any transmitter input greater than 450 pph, but shall not be provided for a transmitter input signal of less than 250 pph. The auxiliary output shall vary proportionally with the excitation input voltage, shall be at the same frequency as the exitation input voltage, and shall be at the same phase within $\pm 0.5^{\circ}$. The auxiliary output signal shall appear across pins 6 and 7 and shall provide fuel flow input to an externally located fuel management subsystem. Pin 6 shall be internally grounded to pin 1. When measuring the auxiliary output voltage, a 200K $\pm$ 1 percent loading resistor shall be placed across pins 6 and 7. These indicators shall meet the test tolerances specified in table I.

3.4.1.2 EFU-22A/A-7 indicators. The requirements for this indicator are similar to those set forth for EFU-22A/A and A-2 except no auxiliary voltage output signal is required. This indicator shall meet the test tolerances specified in table II.

3.4.2 Electrical zero. The synchros in the transmitter and fuel flow indicator shall be so oriented that their electrical zero corresponds to a fuel flow of one-half the full scale value. Identification of the synchro leads and the determination of electrical zero shall be as follows:

a. The rotor leads shall be designated H_2 and C_2, and the C_2 lead shall be grounded.

b. One stator lead shall be designated Z_2 and connected to the C_2 lead.

c. Rated voltage shall be applied across H_2 and C_2, and the voltage measured across H_2 and one of the undesignated stator leads and across the two undesignated stator leads.

d. The rotor shall be rotated until the voltage measured across H_2 and the one undesignated stator lead is approximately a minimum and the voltage across the two undesignated stator leads is exactly minimum. This position is the electrical zero position corresponding to 50 percent of the full-scale value.

e. The rotor shall be rotated in a clockwise direction from electrical zero as viewed from the rotor shaft end, the voltage across one of the undesignated stator leads and Z_2 shall increase, and the voltage across the other undesignated stator lead and Z_2 shall decrease. The stator lead for increasing voltage shall be labeled X_2 and the stator lead for decreasing voltage shall be labeled Y_2.

3.4.3 Power supply. The indicator shall be designed to operate from an aircraft 115V, 400 cycle, single-phase power in accordance with MIL-STD-704.

3.4.3.1 Power consumption. The 115V, 400 Hz power consumption of the indicator (except the type EFU-22A/A-7) shall not exceed 15va, exclusive of the lighting circuit. The power consumption for the type EFU-22A/A-7 shall not exceed 7va. The 5V AC or DC power consumption in the lighting circuit shall not exceed 3va.

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3.4.4 Case. The case shall conform to MS33639 for the 2-inch size, shall be finished in durable dull black, and shall be hermetically sealed. The seal shall be so designed and constructed that it will not be dependent upon materials that might be affected by the action of any atmospheric conditions to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 0.03125 inch. The case length, dimension L, shall not exceed 7.3125 inches, exclusive of the electrical receptacle. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the contracting activity.

3.4.4.1 Filling medium. The filling medium shall be of an least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint - 65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be 12 to 14.7 psi. Other concentrations of helium shall not be used except with prior approval of the contracting activity.

3.4.4.2 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface.

3.4.5 Dials. The dials of the indicator shall conform to figure 1. The numerals and letters shall be in accordance with MS33558.

3.4.6 Counter. A five-digit internal pinion counter shall be used in all types of indicators. The five visible counter numerals appearing in the dial cutout shall all be in the same place and there shall be no discernible difference between the moveable and fixed digits. Ball bearings shall be used to support the main shaft and display wheels. The maximum static counter torque shall not exceed 0.08 inch-ounce at room temperature. A design goal shall be that the counter torque will not exceed 0.2 inch-ounce after 3 million 180-rpm revolutions at an elevated temperature of 75 $\pm$ 5°C. The counters shall have a direct drive and the numbers shall increase when the counter drums are rotated from the top to the bottom. The counters for the EFU-22A/A, A-2 and A/7 indicators shall have the last two counter numerals fixed at zero, and the counter shall indicate the fuel flow rate in 100 pph increments.

3.4.7 Lighting. The indicator dial, pointer, and counter numerals shall be either integrally red lighted in accordance with MIL-L-25467 or integrally white lighted in accordance with MIL-L-27160, as applicable. The internal lighting of the prime standard and all qualification indicators shall be subject to the approval of the qualifying activity.

3.4.7.1 Lamps. Lamps shall be in accordance with MIL-L-25467 or MIL-L-27160, as applicable.

3.4.7.2 Prime standard. The internal lighting of production indicators shall be comparable to and shall balance in brightness and color with a prime standard indicator. The prime standard shall be either an actual indicator conforming to the requirements of this specification or a lighted model. The prime standard shall be used as the reference for acceptance of all production indicators.

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3.4.7.2.1 Prime standard recertification. All representative areas of the face of the prime standard indicator shall be measured for brightness and color as specified in the applicable lighting specification. Each area shall be measured at a sufficient number of locations to obtain data representative of each area including the brightest and dimmest portions of the indicator display. The prime standard indicator shall be measured periodically (at least every 50 hours of operation or every 120 days, whichever occurs first) to determine the magnitude of any brightness or color change. The data and, if required by the qualifying activity, the prime standard shall be submitted for evaluation and reapproval or correction action, or both.

3.4.7.2.2 Visual comparison. The lighting level of the production indicator under test shall fall within the limits of the prime standard when energized at the voltage levels specified in 4.6.26.1.

3.4.8 Pointer. The pointer shall be in accordance with MS33585 except that it shall be suitable for integral lighting as shown on figure 1. The pointer hub and/or pointer tail shall not overlap the counter numerals at any position.

3.4.9 Cover glass. The cover glass shall be clear, flat, and free from flaws which would interfere with the normal reading of the instrument. The glass shall not be fluorescent when exposed to ultraviolet light or to rays in the 363 millimicron region. All reflecting surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-14806.

3.4.9.1 Dial to cover glass distance. The distance between the inner surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.188 inch when indicator display is backlighted, and 0.280 inch when indicator display is wedge lighted. The counter numerals shall be visible when viewed from any direction within a 30° angle from a line perpendicular to the indicator dial at the counter opening.

3.4.10 Compatibility. All EFU-22A/A, /A-2 and /A-7 indicators shall be interchangeable with previously manufactured EFU-22A/A, /A-2 and /A-7 indicators of the same type. The EFU-22A/A, /A-2 and /A-7 indicators shall be completely compatible with TRU-63/A and TRU-63/A-3 transmitters.

3.4.11. Damping. The indicators shall be capable of attenuating small oscillatory transmitter output signals. The EFU-22A/A, /A-2 and /A-7 indicators shall prevent counter responses of more than $\pm$100 pph when a fluctuating input signal of $\pm$1,000 pph at frequencies of 2.2 Hz and higher is applied.

3.4.12 Stops. The indicators shall incorporate internal mechanical stops to limit travel of the pointer and counter so that the pointer travel will be limited to approximately 0.125 inch beyond each end of the scale.

3.4.13 Electrical receptacle. All indicators shall be provided with an MS27034H12B12P hermetically sealed receptacle conforming to MIL-C-26500.

3.4.13.1 Wiring. All indicators shall be wired in accordance with figure 2 or 3 as applicable.

3.4.14 Circuit design. Circuit design shall be such that no electronic components will be operated outside its rated characteristics with 115V $\pm$ 10 percent 400 Hz $\pm$ 10 percent power input and with indicator input signals as

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specified herein. No electronic device shall break down when excited by transients as specified in MIL-STD-704.

3.4.15 Maintainability. The indicator shall be so designed as to provide ease of maintenance. The following requirements shall be included in the design:

a. Rapid and positive identification of the replaceable defective part, assembly or component.

b. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

c. Optimum accessibility of all components requiring maintenance inspections, removals or replacements.

3.4.16. Reliability. The indicators shall have a lower test mean-time-between-failure (MTBF) (0_1) as defined in MIL-STD-781, demonstrated to a 90% confidence level as follows:

Type	0_1 (hours)
EFU-22A/A & /A-2	1,200
EFU-22A/A-7	2,000

3.4.17 Longevity. The indicator shall have an operation life span before wearout failures occur or the indicator consistently fails to meet the lower MTBF index as follows:

Type	Operating Life Span (Hours)
EFU-22A/A & /A-2	Not less than 2,500
EFU-22A/A-7	Not less than 4,000

Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the contracting activity.

3.4.18 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.5 Performance. The indicators shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperature - temperatures ranging from -54° to +71°C; except the EFU-22A/A-7 indicator shall be -40°C to +71°C and storage temperature ranging from -62°C to +71°C; except the EFU-22A/A-7 shall be -54°C to +95°C.

b. Vibration - vibration as specified in paragraph 4.6.17.

c. Thermal shock - eight cycles of alternate immersions in water at 85° and 5°C for 30 minutes.

d. Salt fog - salt fog for a period of 50 hours.

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e. Voltage and frequency variation - voltage and frequency variation ranging from 110V 380 Hz to 120V 420 Hz as specified in 4.6.10.

f. Voltage surge - application of 180V for 0.09 seconds.

3.5.1 Fogging. There shall be no fogging of the face of the indicator glass when the indicator is energized and kept in a $70^{\circ} \pm 2^{\circ}$C controlled ambient temperature for a minimum of 1 hour and the glass face then rubbed with an ice cube for a period of 1 to 2 minutes.

3.5.2 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for Class A-1 equipment.

3.5.3 Magnetic effect. The indicators, operating or not operating, shall not create a magnetic field with an intensity that will deflect a short bar magnetic compass more than 2° when tested in accordance with paragraph 4.6.20.

3.5.4 Response. The EFU-22A/A and EFU-22A/A-2 indicators shall show a rate indication from zero to full scale or from full scale to zero in less than 25 seconds under room temperature and high and low ambient conditions. The EFU-22A/A-7 indicator shall respond to a step input of 70,000 pph within 6 seconds.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Weight. The weight of the indicators shall not exceed 1.5 pounds.

3.8 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.8.1 Nameplate. A nameplate shall be provided on the rear of the case. When applicable, the information shall conform to MIL-STD-130. Metal decalcomanias shall not be used.

3.8.2 Electrical schematic. An electrical schematic showing the function of each connector pin shall be permanently attached to the indicator. An approved metal decalcomania may be used provided it is permanently attached and cannot be peeled back with the fingernail.

3.9 Workmanship. The indicators shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operations. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable

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to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

"4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. First article inspection

b. Acceptance tests

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.2 Vibration. Unless otherwise specified, the indicators shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Test position. Unless otherwise specified, the indicator shall be tested with the case in a horizontal position with the scale upright.

4.3.4 Master transmitter. The indicator shall be tested with a master transmitter that generates a signal in accordance with 3.5.1. Sufficient tests shall be performed to determine that the transmitter signal is not in error in excess of 0.03 percent.

4.3.6 Tolerances. Test tolerances shall be as specified in tables I and II.

4.3.7 Power supply. Unless otherwise specified, the indicators under test shall be energized with 115 $\pm$ 1V, 400 $\pm$ 1 Hz power. The lighting circuit shall be energized with 4.75 $\pm$ 0.25V, 400 $\pm$ 1 Hz power.

4.3.8 Test equipment. The test equipment shall not load nor affect the normal operation of the indicator. A typical test setup for the indicators is shown on figure 4. The following test instruments, or their equivalents, may be used:

a. Rotary test head, Millitest Corporation model THM 101 (with front and rear flange size 11 synchro adapter)

b. Master synchro transmitter, Clifton model HGC-11-AR-6/A758

c. Phase angle voltmeter, North Atlantic model No. 202BR

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d. Isolation transformer, Gretsch model No. ST-248

e. Geoformer transformer, Triad model No. G-40

f. Ratio control box, North Atlantic model No. RB-504

4.4 First article inspection.

4.4.1 Test samples. The first article test samples shall consist of at least seven indicators representative of the production equipment. Three of the samples shall be subjected to all the tests in sampling plan B, and seven samples (may include the three subjected to the sampling plan B, test) shall be subjected to the sampling Plan C tests. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.4.2 First article tests. The first article tests shall consist of all the test specified under 4.6. Unless otherwise specified by the qualifying activity, the tests shall be conducted in the order listed.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Voltage surge

c. Scale error

d. Friction

e. Hysteresis

f. Pointer error

g. Response

h. Sealing

i. Burn-in

j. Fogging

k. Lighting (Individual tests of MIL-L-25467 or MIL-L-27160, as applicable)

l. Damping

4.5.2 Sampling tests.

4.5.2.1 Sampling plan A. A minimum of 3 indicators shall be selected at random from each 100 or fraction thereof produced on each contract and subjected to the following tests as described under 4.6:

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a. Individual tests

b. Vibration

c. Voltage and frequency variation

d. Low temperature

e. high temperature

f. Position error

g. Lighting (Sampling plan A tests of MIL-L-25467 or MIL-L-27160, as applicable)

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 10 items produced on each contract or order and subjected to the following tests in the order specified and as described under 4.6.

a. Sampling plan A tests

b. Magnetic effect

c. Low temperature exposure

d. High temperture exposure

e. Salt fog

f. Thermal shock

g. Vibration endurance

h. Elecromagnetic interference

i. Lighting (Sampling plan B tests of MIL-L-25467 or MIL-L-27160, as applicable)

4.4.2.2.1 Test schedule. When sampling plan B tests are not specifically waived on a contract, the test shall begin as soon as the first 10 items have been produced but before any production deliveries have been made. The test data will be submitted to the contracting activity no later than 16 weeks after starting the test program.

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3.4 Sampling plan C. Seven indicators shall be selected at random from the first 14 produced and subjected to the following tests as described under 4.6. Rejection shall be in accordance with 4.6.24.6.

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a. Individual tests

b. Reliability

c. Longevity

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, marking and dimensions. The indicator shall also be weighed to determine compliance with paragraph 3.8.

4.6.2 Voltage surge. Power of 102 ± 1V, 360 ± Hz shall be applied for 10 minutes followed immediately by application of 124 ± 1V, 440 ± 5 Hz for 10 minutes. The indication shall be 20,000 pph during this test. At the end of the second 10 minute period, the voltage and frequency shall be adjusted to 115 ± 1V, 400 ± 5 Hz. A voltage surge shall then be applied to the power input to provide a momentary root mean square (rms) voltage of 180 ± 2V and return to 115 ± 1V. The duration of the surge, including rise and decay time, shall be 0.09 ± 0.005 seconds. At the end of this test, the scale error test shall be conducted with tolerances as specified at room temperature. Erratic operation or failure of the scale error test shall consititute failure of the voltage surge test.

4.6.3 Scale error and friction.

4.6.3.1 Scale error. The indicator shall be connected to a master transmitter and to test equipment that will measure the auxiliary output, when applicable, to within 0.01 percent. The master transmitter shall be set at angles equivalent to each of the test points in the applicable scale error test tolerance table. The indicator counter and output tolerances (where applicable) shall be within those specified therein. The indicator shall be vibrated before each reading is taken. Prior to conducting this test, the indicator shall be driven at maximum speed against both the upper and lower stops. Eight consecutive cycles shall be completed before the accuracy check is performed. The specific point at which the voltage output can be detected and goes to zero for both the upscale and downscale directions, respectively, shall also be recorded after each accuracy check. The master transmitter shall then be set at zero and slowly increased. The auxiliary output signal shall be zero until the input signal reaches the applicable turn-on level specified in 3.5.1. The auxiliary output as measured across pins 6 and 7 shall then be in accordance with the applicable test tolerance table. The applicable turn-off level specified in 3.5.1 shall be approached slowly from the opposite direction. Within the specified tolerances, the auxiliary output signal shall go to zero and remain at zero through the remaining signal range to zero pph.

4.6.3.2 Friction. The indicator shall be tested for friction at the test points specified in the applicable test tolerance table. The master transmitter shall be made to slowly approach the test point in an ascending direction. After the transmitter had been held at the test point for approximately 5 seconds, the

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indicator pointer reading shall be recorded. The indicator shall then be gently vibrated to establish a stable condition and the reading shall be recorded. The difference between the two readings shall be recorded as friction and shall not exceed the specified tolerances.

4.6.4 Hystersis. The master transmitter shall be made to slowly approach the test point shown in the applicable test tolerance table and shall be vibrated. Readings shall be taken at the test point when it has been aproached in this manner from both the upscale and downscale directions. The difference in these readings shall not exceed the tolerances specified.

4.6.5 Pointer error. The master transmitter shall be set so that the pointer is exactly in the center of the graduztion identified by the test point shown in the applicable test tolerance table. The counter and output voltage where applicable shall read the same corresponding value of fuel flow within the tolerance specified therein.

4.6.6 Response. To test the EFU-22A/A and /A-2 indicators, the master transmitter setting shall be changed instantaneously from a value corresponding to a flow rate of 0 pph to a value corresponding to the full-scale flow-rate value. The indicator shall read within 1% of this value in less than 25 seconds.

4.6.6.1 Step input response. The EFU-22A/A-7 indicator shall be tested by introducing a step input change from 0 pph to 70,000 pph. The indicator shall read within 7,000 pph of 70,000 pph for all time after 6 seconds of initiation of the step input. The EFU-22A/A-7 shall be similarly tested with a downscale step input change from 80,000 pph to 10,000 pph. The same 6 second criteria shall apply.

4.6.7 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmoshpere.

4.6.8 Burn-in. The indicator shall be placed in a chamber and the temperature maintained at 50°C for minimum of 24 hours. Power shall be applied in accordance with 4.3.7 throughout the test while maintaining a constant signal input equivalent to a half-scale indication. At least 8 hours of this test shall be performed during normal working hours, and observations shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the indicator display or the output auxiliary voltage change by more than one-half of the tolerance specified for that test point at room temperature in the applicable test tolerance. Pointer or counter oscillation or erratic movement shall be cause for rejection. The indicator shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.3.1.

4.6.9 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514.2, category b.1, curve B. The indicator shall be operating and shall be vibrated in each of the three mutually perpendicular planes. During this test, pointer, and output voltage and variation shall not exceed the tolerances specified in the applicable test tolerance table. Following the vibration period, the indicator shall meet the tests specified in 4.6.3.1, 4.6.3.2, and 4.6.4. Room temperature tolerances shall not be exceeded and screws and other parts shall not become loosened nor damaged as a result of the applied vibration.

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4.6.10 Voltage and frequency variation. The master transmitter shall be set to the test point listed in the applicable test tolerance table. The voltage and frequency applied to the master transmitter and indicator shall be as follows. When the applied voltage and frequency are changed as specified from 115V, 400 Hz, the change in indication shall not exceed the tolerance specified.

a. 110V ac, 380 Hz

b. 110V ac, 420 Hz

c. 120V ac, 380 Hz

d. 120V ac, 420 Hz

4.6.11 Low temperature. The EFU-22A/A or /A-2 indicator, not energized, shall be subjected to a temperature of $-54 \pm 2^{\circ}C$ for a period of 4 hours. While still at this temperature, the transmitter shall be set at zero pph. Power shall be applied to the indicator. At the end of a 10-minute period, the master transmitter shall be set at the full-scale test point. Within 180 seconds, the indicator shall read within the scale error tolerance for this point listed in the applicable test tolerance table. The indicator shall then meet the low temperature scale error tolerances specified in the applicable table for all the test points. At the conclusion of this test and while the indicator is still in a $-54^{\circ}C$ ambient, the indicator shall meet the test specified in 4.6.6. Response time shall not exceed 25 seconds from zero to full scale indicator. The test shall be conducted immediately after the low temperature scale error test and prior to power being applied to the indicator for an accumulation of 40 minutes. The EFU-22A/A-7 indicator, not energized, shall be subjected to a temperature of $-40 \pm 2^{\circ}C$ for a period of 4 hours. While still at this temperature, the test transmitter shall be set at zero pph and power shall then be applied to the indicator for a period of one minute. At the end of the one minute, a step input signal of 2,000 pph shall be applied to the indicator. The indicator shall read 1,800 pph within 20 seconds of initiation of the step input. While the indicator is still at a $-40^{\circ}C$ ambient and within 240 seconds of initial power application, the indicator shall meet the upscale portion of the response test specified in paragraph 4.6.6. The response time shall not exceed 25 seconds for the 70,000 pph step change. The downscale portion of the test of paragraph 4.6.6 shall then be performed with the same 25 second response time tolerance. The indicator shall then meet the low temperature scale error tolerances specified in the applicable table for all the test points.

4.6.12 High temperature. The EFU-22A/A or /A-2 indicator, while energized, shall be subjected to a temperature of $71 \pm 2^{\circ}C$ for a period of 4 hours, and tested at this temperature for scale errors. The scale errors shall not exceed the tolerances specified in the applicable test tolerance table. At the conclusion of this test and while the indicator is still in a $71^{\circ}C$ ambient, the indicator shall meet the test specified in 4.6.6. The response time shall not exceed 25 seconds from zero to full-scale indication. This test shall be conducted immediately after the test specified in 4.6.3 at high temperature. The EFU-22A/A-7 indicator, while energized, shall be subjected to a temperature of $71 \pm 2^{\circ}C$ for a period of 4 hours, and tested at this temperature for scale errors. The scale errors shall not exceed the tolerances specified in the applicable test tolerance table. At the conclusion of this test and while the indicator is still in a $71^{\circ}C$ ambient, the indicator shall meet the test specified in 4.6.6. The response time shall not exceed 12 seconds for the 70,000 pph step change. This test shall be conducted immediately after the test specified in 4.6.3 at high temperature.

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4.6.13 Position error. The test specified in 4.6.3 shall be repeated at the test points shown in the applicable test tolerance table with the instrument turned 90° right, 90° left, face up and face down. The change in scale error shall not exceed the tolerances specified therein.

4.6.14 Damping. The input signal to the indicator shall be made to oscillate about the test points specified in the applicable test tolerance table at a frequency of 2.2 Hz with the double amplitude as specified in 3.5.11. Oscillation of the counter, pointer, and auxiliary output (if used) shall not exceed a total excursion of ± 100 pph.

4.6.15 Fogging. The indicator shall be energized and kept in a 70 ± 2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at the high temperature, and ice cube shall be rubbed on the face of the indicator glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water or oil fog. Any evidence of fogging shall be cause for rejection. This test may be conducted in conjuction with the test specified in 4.6.3.1 and 4.6.3.2 at high temperature.

4.6.16 Thermal shock. The indicator shall be subjected to a total of 8 cycles of immersion in tap water at a temperature of 85 ± 2°C and then in tap water at a temperature of 5 ± 2°C. The length of time for each bath immersion shall be 30 minutes with not more than 3 seconds between immersions. The indicator shall then pass the test specified in 4.6.7.

4.6.17 Vibration endurance. While operating, the EFU-22A/A or /A-2 indicator shall be vibrated in each or the three mutual perpendicular planes within a range of 5 to 500 Hz for a period of 6 hours in each plane. Vibration shall be applied at a double aplitude of 0.010 inch or at an acceleration of ±1.5g, whichever is the lower value. The rate of frequency cycling shall be logarithmic and such that 15 minutes will be required to proceed from 5 to 500 Hz and return to 5 Hz. The indicator shall be set at one-half of full scale test point. When there is no provision for logarithmic cycling, a linear rate of frequency may be used. Following the vibration period, the indicator shall meet the tests specified in 4.6.3.1, 4.6.3.2 and 4.6.4.

4.6.17.1 Dwell. The EFU-22A/A-7 indicator shall be subjected to vibration in accordance with MIL-STD-810, Method 514.2, equipment category b.1, curve J. In addition to the resonant dwells found and specified by Method 514.2, the indicator shall be subjected to dwells at 100 ± 10 Hz, 200 ± 20 Hz, 300 ± 30 Hz, and 400 ± 40 Hz. These dwells shall be for 5 minutes in each of the three mutually perpendicular indicator axes and shall be at a 5g acceleration level. The indicator shall be powered and indicate 20,000 pph throughout the test. The indicator shall be subjected to and meet the requirements of the tests specified in 4.6.3.1, 4.6.3.2, and 4.6.4 following the vibration.

4.6.18 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 502.1, except after the 48-hour exposure at -62°C, the temperature shall be -54°C and maintained for 24 hours (-40° for EFU-22A/A-7 indicator). Upon completion of the 24-hour period at -54°C (-40° for the EFU-22A/A-7 indicator), and while still at this temperature, the Type EFU-22A/A and /A-2 indicators shall be subjected to the test specified in 4.6.3 and shall meet the tolerances specified in 4.6.11. The Type EFU-22A/A-7 indicator temperature shall be raised to -40°C for 4 hours to testing. The temperature shall

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be raised to room temperature and shall be maintained for a period of not less than 4 hours and the indicator subjected to the test specified in 4.6.3 at room temperature. The scale errors shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to extreme temperature exposure.

4.6.19 High temperature exposure. The high temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 501.1, Procedure 1. After the 48-hour period at +71°C and while at this temperature, the indicator shall meet the test specified in 4.6.3. The position error test shall be conducted and the errors shall not exceed those specified in 4.6.13. The indicator shall be maintained at room temperature for a period of not less than 4 hours after which it shall meet the test specified in 4.6.3 at room temperature. The errors osbserved shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to the extreme high temperature exposure.

4.6.20 Magnetic effect. The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.5 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 135°, 180°, 225°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 2°. The test shall be repeated with the indicator operating at rated voltage.

4.6.21 Salt fog. The indicator, or a hermetically sealed mockup that exactly duplicates the external portion of the indicator, shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1 for a period of 50 hours.

4.6.22 Electromagnetic interference. The indicator shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, test methods CE03, CS01, CS02, CS06,RE02, RS02, and RS03. For first article purposes, this test need be conducted on only one indicator.

4.6.23 Compatibility. Compatibility tests (see 6.2) shall be conducted as required to determine compliance with previously procured test equipment (AGE).

4.6.24 Reliability. The seven indicators specified in 4.5.2.4 shall be subjected to a reliability test to demonstrate a lower test MTBF (0_1) at a 90% confidence level as follows:

TYPE	0_1 Hours
EFU-22A/A & /A-2	1,200
EFU-22A/A-7	2,000

The test shall be conducted in accordance with MIL-STD-781, test plan IC.

4.6.24.1 Test equipment. The test equipment and fixtures shall include provisions for indicator operation cycling, temperature cycling, and vibration.

4.6.24.2 Operation cycling. During this test, the indicator shall be continously cycled from 500 pph to 90 percent of full scale and return at a rate of 1 cycle every 4 minutes.

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4.6.24.3 Temperature cycling. Unless otherwise specified or amended by the results of the thermal analysis, the temperature of the chamber in which the indicators are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperature to -54°C for Type EFU-22A/A and EFU-22A/A-7. -40°C shall be the minimum for Type EFU-22A/A-7. Stabilize for 1.5 hours with indicators unenergized.

b. Turn indicators on and raise temperature to 55°C and operate indicators for 4 hours.

c. Turn indicators off and lower chamber temperature to -54°C for Type EFU-22A/A and EFU-22A/A-2 for 2 hours. The minimum temperature for Type EFU-22A/A-7 shall be -40°C.

d. Repeat steps b and c until reliability requirement has been met. This schedule provides four 6-hour test cycles for each 24-hour period. In each 24-hour period, the operational test time is 16 hours.

e. Minimum rate of temperature change shall be 5°C per minute.

4.6.24.4 Vibration cycling. Throughout this test, the indicators shall be vibrated for a period of 10 minutes during each hour of on-time at any noresonant frequency between 20 and 60 Hz and a peak acceleration of 2.2g ± 10%.

4.6.24.5 Performance check. At each 200-hour interval and at the end of this test, the indicators shall be subjected to the following performance tests: scale error, friction, hysteresis, and response time. Lighting data (brightness and color tests) shall be taken at 200-hour intervals for information purposes only. During each scale error test, if any two test points exceed the allowable temperature-endurance tolerances specified in table I, or II as applicable, one failure shall be charged against this test.

4.6.24.6 Failure criteria. For an accept decision, each EFU-22A/A and /A-2 indicator shall be operated at least 1,000 hours. Each EFU-22A/A-7 indicator shall be required to operate at least 1,700 hours. If an indicator fails, entry shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the indicators still under test. The contracting activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, paragraph entitled "accept-reject criteria for: Test Plan IC".

4.6.25 Longevity. Three of the seven reliability test samples shall continue to be tested through a longevity test until 2500 operating hours for Type EFU-22A/A and EFU-22A/A-2 (4000 operating hours for Type EFU-22A/A-7) have accumulated on each of the three indicators. The equipment shall be tested under the enviromental conditions specified herein. Test time accumulated on the equipment during a reliability test may be applied toward the longevity test when the same sample units are used.

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4.6.25.1 Evaluation of longevity test. No accept/reject criteria for this test is specified. The evaluation shall be limited to the analysis of all faliures, indentification of design and pattern failures, and the submission of recommended design changes to the contracting activity.

4.6.26 Visual comparison. The following test shall be conducted during the individual production quality conformance testing. The test shall be conducted by qualified personnel only after they have been previously tested for normal color vision and passed and have become dark adapted prior to performing any visual comparison tests. The prime standard and the indicator undertest shall be placed side-by-side at eye level in completely dark surroundings and viewed at a distance of approximately 2.5 feet in a sighting cone 45° from the normal to the surface of the cover glass. All areas of the indicator including markings, pointers, flags, et cetera, shall be compared when viewed from any position within the sighting cone.

4.6.26.1 Brightness. With 2.50 $\pm$ 0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the standard, like areas of the test indicator and standard shall be compared as: pointers, command bars, alarm flags, numerals, indicia, markings, et cetera for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness levels of the prime standard when set at 2.40 and 2.60V. If the brightness levels do not meet this criteria in any individual area or the relative uniformity of lighting of the indicator and prime standard are different, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this photometric test, they shall be rejected and returned for rework. First article units shall not be tested by use of the prime standard. The first article units must be photometrically measured in accordance with the applicable lighting specification.

4.6.26.2 Color. A color comparison test shall be conducted with 4.50V applied to both the production indicator and the prime standard. The two indicators shall appear indentical in color. Questionable areas of the production units shall be measured in accordance with MIL-L-25467 or MIL-L-27160.

4.6.27 Effect of mechanical stops. The indicator shall be driven at its maximum response speed against both the upper and lower stops. This shall constitute one cycle. Eight consecutive cycles shall be repeated and the scale error and friction test of 4.6.3.1 and 4.6.3.2 shall then be conducted.

4.7 Inspection packaging. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5 .1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with method 1A8 of MIL-P-116. Each indicator shall be over-boxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Indicators shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified.

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indication both main engine and afterburner flow rate.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of indicator required (see 1.2)

c. When sampling plan B test will not conducted (see 4.5.2.2)

d. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense

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e. Procedures for conducting compatibility tests (see 4.6.23)

f. Levels of packaging and packing required (see 5.)

g. When a first article is required (see 3.1 and 4.4.2)

6.3 Definition.

6.3.1 Hermetic seal. A hermetic seal is a perfectly closed, airtight seal made between vitric or metallic materials, or both. A hermitic seal is not intended to include seals accomplished by gaskets or "O" rings.

6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should issue specific instructions in aquisition documents regarding arrangements for examinations, test approval of the first article.

6.4.1 Approval of first article test procedures. The first article test procedures will be submitted for formal approval before the test program is initiated.

6.4.2 Disposition of first article test spamles. Unless otherwise specified, the first article test samples will be retained by the qualifying activity.

6.5 Subject term (key word) listing.

Fuel
Indicator
International interest
Rate of Flow
Syncho

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ASCC Air Standard 10/69. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriated action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accomodations.

6.7 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force 71

Review activity:
Air Force - 11

Project number:
6620-F367

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TABLE I. Test tolerances for EFU-22A/A and /A-2 indicators.

TEST	TEST POINT (PPH)	COUNTER SCALE ERROR TOLERANCES (±PPH)		OUTPUT SIGNAL (VOLTS)	OUTPUT SIGNAL TOLERANCE (±MV)		
		ROOM TEMPERATURE	±71° AND -54°C AND AFTER ENDURANCE		ROOM TEMPERATURE	-54°C	+71°C AND AFTER ENDURANCE
Scale Error	0	100	200	0	–	–	–
	500	100	200	0.0625	9	28	23
	1,000	100	200	0.125	9	28	23
	2,000	100	200	0.250	8	27	22
	4,000	100	200	0.500	6	25	20
	6,000	100	200	0.750	8	27	22
	9,000	100	200	1.125	12	31	26
	12,000	100	200	1.500	14	33	28
	15,000	100	200	1.875	16	35	30
	20,000	100	200	2.500	18	37	32
	25,000	100	200	3.125	20	39	34
	30,000	100	200	3.750	23	42	37
	40,000	100	200	5.000	26	45	40
	50,000	100	200	6.250	29	48	43
	60,000	100	200	7.500	33	52	47
	70,000	100	200	8.750	35	54	49
	80,000	100	200	10.000	37	56	51

TEST	TEST POINT (PPH)	TOLERANCES (±PPH)		OUTPUT SIGNAL TOLERANCE (±MV)
		COUNTER	POINTER	
Hysteresis & Damping	0	100	500	0
	4,000	100	500	15
	9,000	100	500	30
	40,000	100	500	65
	80,000	100	500	92
Voltage & Frequency Variation	4,000	100	500	28
Vibration	4,000	100	500	15
Pointer Error	4,000	100	500	15
	40,000	100	500	65
Position Error	4,000	100	500	15
Friction	4,000	100	500	15
	9,000	100	500	30
	40,000	100	500	65
	80,000	100	500	92

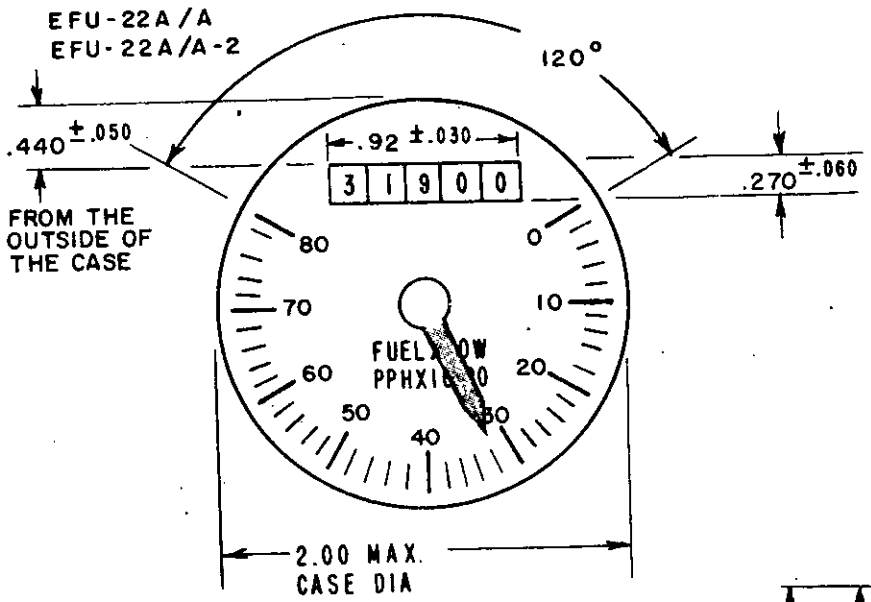
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TABLE II. Test tolerances for EFU-22A/A-7 indicator.

TEST	TEST POINT (PPH)	COUNTER ERROR TOLERANCES(±PPH)	
		ROOM TEMPERATURE	+71° C AND AFTER ENDURANCE
Scale Error	0	200	300
	500	200	300
	1,000	100	200
	2,000	100	200
	4,000	100	200
	6,000	100	200
	9,000	100	200
	12,000	100	200
	15,000	100	200
	20,000	100	200
	25,000	100	200
	30,000	100	200
	40,000	100	200
	50,000	100	200
	60,000	300	400
	70,000	350	450
	80,000	350	450

TEST	TEST POINT (PPH)	TOLERANCES (±PPH)	
		COUNTER	POINTER
Hysteresis & Damping	0	100	500
	4,000	100	500
	9,000	100	500
	40,000	100	500
	80,000	100	500
Voltage & Frequency Variation	4,000	100	500
Vibration	4,000	100	500
Pointer Error	4,000	N/A	500
	40,000	N/A	500
Position Error	4,000	100	500
Friction	4,000	100	500
	9,000	100	500
	40,000	100	500
	80,000	100	500

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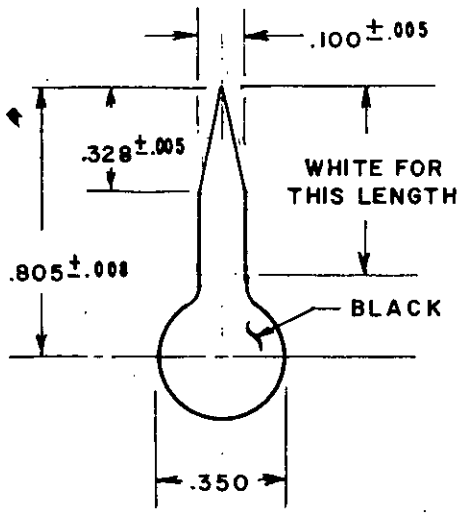


Dial Dimensions

Markings	Height	Width
Scale Numerals	0.156	0.020
Major Graduations	0.156	0.030
Minor Graduations	0.093	0.015
FUEL FLOW	0.093	0.02
PPH x 1000	0.062	0.015
Counter Numerals	0.190	0.025
500 pph graduation	0.062	0.015

Dimensions in inches.

Tolerances: Height ± .0156
Width ±.005



POINTER DIMENSIONS

DIAL COLOR
ALL INDICIA: WHITE COLOR NO 37875 FED-STD-595
ALL BACKGROUND: BLACK COLOR NO. 37038 FED-STD-595

FIGURE 1. Indicator presentation.

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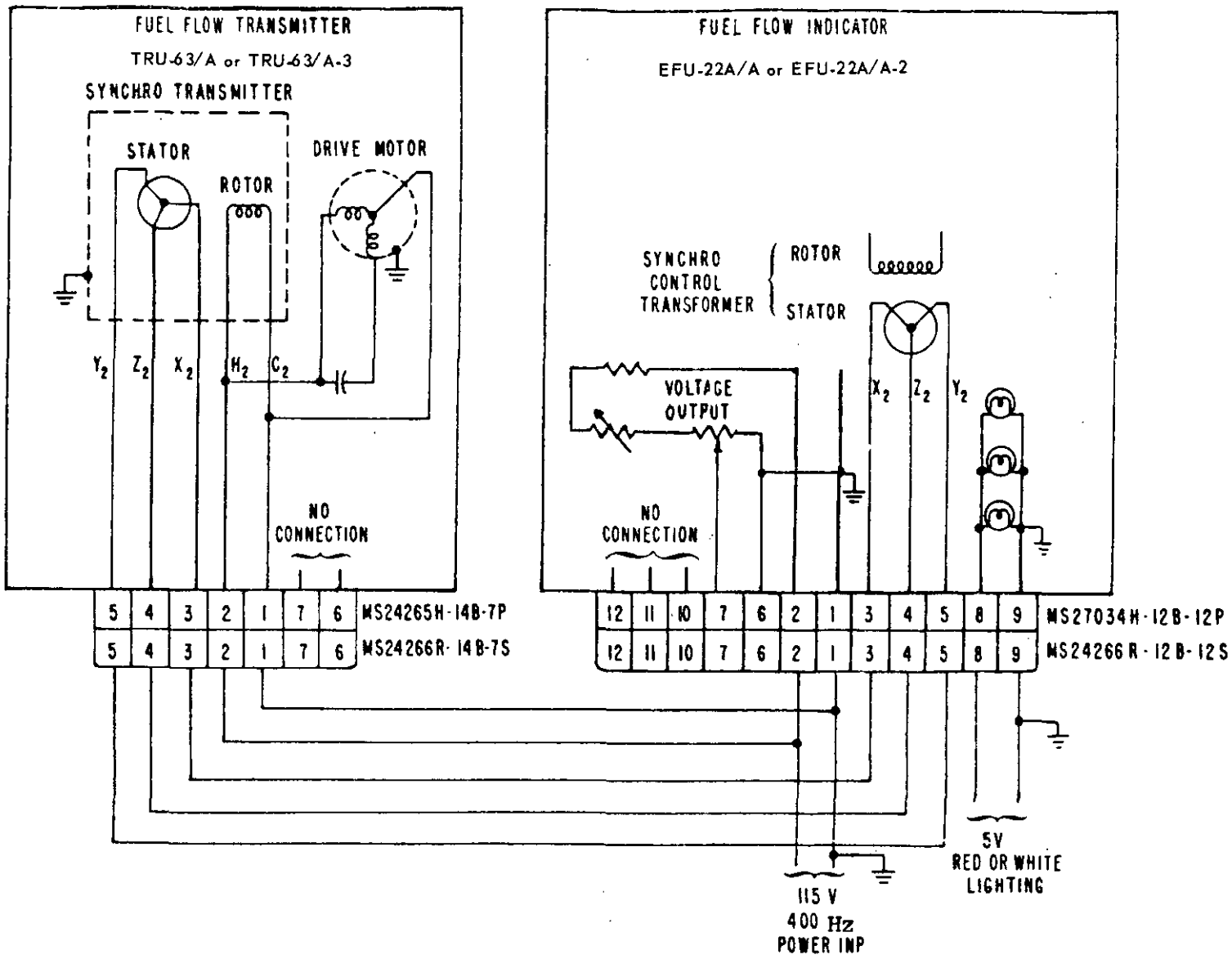


FIGURE 2, System wiring diagram.

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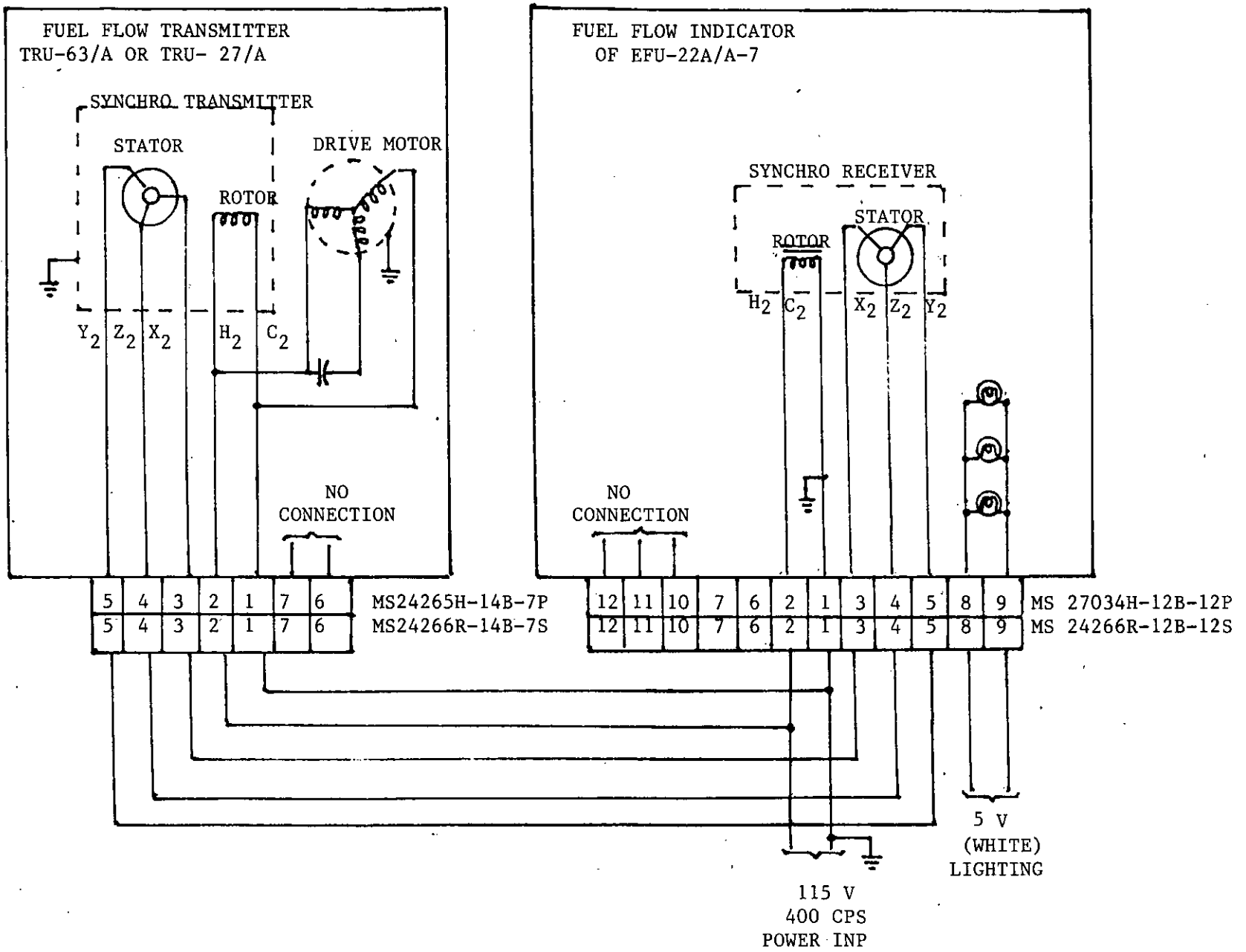


FIGURE 3. System wiring diagram.

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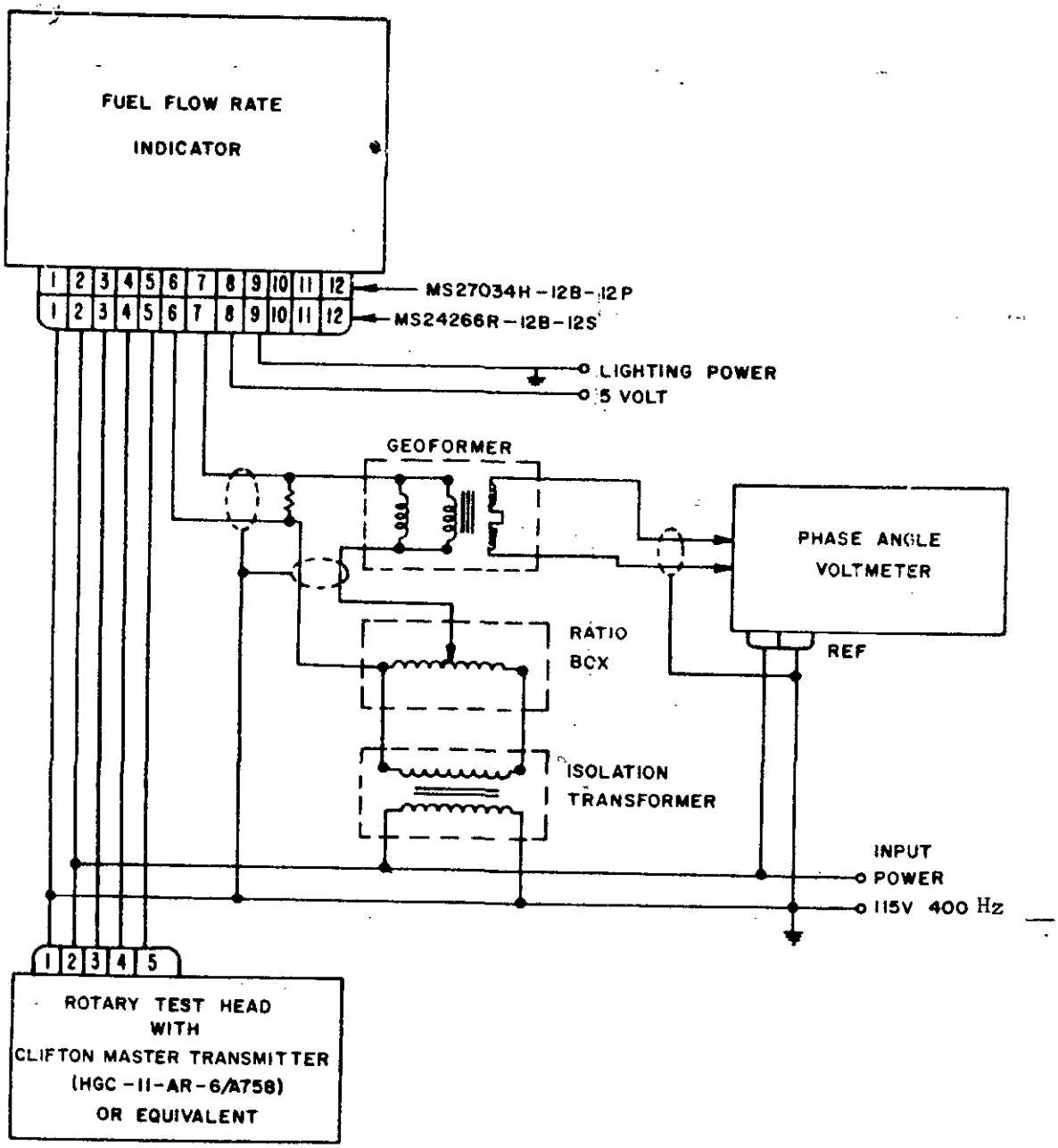


FIGURE 4. Typical test setup.

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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